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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F)

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TITLE OF THE PROJECT	CodeLens — Code Understanding & Impact Analysis	
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Abstract

Software systems continuously evolve through bug fixes, feature additions, optimizations, and maintenance. Even a small modification in an existing codebase can affect multiple functions, variables, modules, or dependencies, making it difficult for developers to accurately determine the consequences of a change. Understanding unfamiliar source code manually is also time-consuming and error-prone, especially when the codebase grows in size and complexity. **CodeLens — Code Understanding & Impact Analysis** is a compiler-based system designed to address these challenges by analyzing existing C source code and providing structured information about its program structure, relationships, and potential change impact.

CodeLens focuses on **non-AI, static analysis** using fundamental compiler design techniques. The system processes C source code through multiple analysis stages, beginning with lexical and syntactic analysis to identify program constructs such as keywords, identifiers, operators, functions, variables, and control structures. The extracted information is organized into suitable internal representations, including symbol information and relationships between program entities. The system then performs dependency and structural analysis to identify how functions, variables, and other components are connected within the source code.

A key objective of CodeLens is **change impact analysis**. When a developer modifies a particular function, variable, or other program element, the system traces the relevant dependencies and identifies the portions of the code that may be affected by the modification. This provides developers with a clearer understanding of the possible consequences of a change before they manually inspect the entire codebase. The analysis can help identify directly dependent components as well as related elements that may require further testing or review.

The proposed system follows a structured workflow consisting of **source-code input, lexical analysis, parsing, intermediate representation, symbol and dependency analysis, and impact analysis**. The resulting information is presented in an understandable form so that developers can explore the structure of

the code and determine which components are associated with a selected program element. By relying on deterministic compiler-based analysis rather than artificial intelligence, CodeLens aims to provide explainable and reproducible results based directly on the source code and its relationships.

CodeLens demonstrates how concepts from compiler design can be applied to a practical software-engineering problem beyond traditional compilation. The system can serve as a foundation for understanding legacy or unfamiliar C programs, supporting maintenance activities, and reducing the effort required to assess code changes. Ultimately, the project aims to bridge the gap between **compiler-level program analysis and practical software maintenance** by providing a systematic approach to code understanding and change impact analysis.

Signature of the Guide

HOD